Percentage

Assignment - I

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1. 20% of ? = 90

2. $6\frac{1}{4}$% of ? = 5400

3. 26% of x = 2564 – 2226

4. $2\frac{1}{4} \times 24 =$ x% of 360

5. 70% of 350 = x% of 490

6. 18 is what % of 144?

7. $\frac{7}{20}$ is what % of $\frac{7}{10}$?

8. 25% of what is $\frac{1}{9}$?

9. 35% of what is 70×8

10. 490% of 25 + 25% of 490 = ?

11. 40% of 1020 + ?% = 70% of 990 + 300% of 425

12. 600% of 1600 = ?% x 1200

13. 25% of (250% of Rs.800) = ?

14. If Y% of Y is 49, then Y is equal to__.

15. What is 25% of 25% equal to?

16. What is 60 percent of Rs.720?

17. What % of 500 gm is 2 Kg?

18. Two numbers are respectively 10% and 25% more than a third number. First number is how much % less than the second number?

19. Two numbers are respectively 40% and 60% of the third number. What % of the first number is the second number?

20. Two numbers are respectively 20% and 30% less than a third number. What % of first number is the second number?

21. A person gives 30% of his salary to his wife and 45% of his salary to his children. Now, he is left with Rs. 5,000. Find his total salary?

22. A person gives 20% of his salary to his wife, 15% of his salary to his son and 25% of his salary to his daughter. Now, he is left with Rs. 16,000, What is his total salary?

23. A person gives 30% of his salary to his wife and 20% of the remaining to his children. Now he is left with Rs. 28,000. What is his total salary?

24. A person gives 20% of his salary to his wife and 25% of the remaining to his children. Now, he is left with Rs. 27,000. What is his total salary?

25. In an examination, it is required to get 45% of the total marks to qualify. Ramesh got 590 marks and was declared failed by 40 marks. What are the maximum marks a student can get?

26. In an examination, it is required to get 55% of the total marks to qualify. Suresh got 500 marks which was 5 marks more than the qualifying marks. What are the maximum marks a student can get?

27. Price of a toy was increased by 40% and then decreased by 25%. What was the net change in the price (in %)?

28. Rohan gets 35% of the marks in an examination and fails by 75 marks. Sohan gets 540 marks in the same examination and fails by 60 marks. Find the maximum marks?

29. A candidate who got 24% in a test failed by 45 marks. Whereas another candidates got 240 marks in the same test, which was 15 more than the passing marks, find the maximum marks?

30. In an examination, 25% of total students failed in Hindi, 40% failed in English and 15% in both. Find the percentage of those who passed in both the subjects.

31. What is the difference between $\frac{1}{4}$th of 2000 and $\frac{1}{4}$th % of 2000?

32. In a college of 15,000 students, 30% of students participated in a game show. Out of all those who participated, 25% are girls. Find the number of boys participating in the show?

33. In a college of 1500 students, 400 are in Arts stream, 600 are in Science stream and remaining are in Commerce stream. What is the % of students studying Science?

34. The sum of 25% of a number and 12% of the same number is 2590. What is 15% of that number .

35. $\frac{1}{5}$th of $\frac{2}{5}$th of $\frac{3}{5}$th of a number is 288. What is 25% of that number?

36. If the height of a triangle is decreased by 10% and base of the same is increased by 20%. What will be the effect on its area?

37. There are 1500 employees in an office. Number of females is 25% of the number of males. Find the total number of females in the office?

38. If the length of a rectangle is increased by 10% and breadth is reduced by 20%, What will be the effect on its area?

39. If the price of Petrol increases by 10%. By how much % a person should decrease his consumption, so that his expenditure on petrol does not increase.

40. If the price of Petrol decreases by 10%. By how much % a person should increase his consumption, so that his expenditure on petrol does not decrease.

41. If the price of rice increase by 20%. By how much % a person should decrease his consumption, so that his expenditure on rice does not increase.

42. If a number is added to 21 instead of being added to 12. What % of the correct value is the result?

43. If a number is added to 20 instead of being added to 120. The correct value is how much % more than the result?

44.What % of number of days in May 2008 is number of days in Feb.2008.

45. By how much % number of days in a week is less than number of hours in a day?

46. A shopkeeper marks the price of a television 25% higher than the original price and them allows a discount of 25%. What profit / loss % he will get in the transaction?

47. A shopkeeper marks the price of a mobile 20% higher than the original price and them allows a discount of 15%. What profit / loss % he will get in the transaction?

48. Shaila decided to donate 15% of her salary to an NGO. On the day of donation, she changed her mind and donated Rs. 2,475 Which was 75% of what she decided earlier. What is Shaila's total salary?

49. There are 8200 employees in a company, out of which 65 per cent are males and 60 per cent of the males are either 40 years or older. How many females in the company are younger than 40 years?

50. In an Examination, Pravin scored 49 per cent marks, Rahat scored 98 per cent marks and Sonali scored 825 marks. The maximum marks of the examination are 900. What are the average marks scored by all the three students together?

Solutions:-

1. 20% of ? = 90

?/5 = 90

? = 450

2. $6\frac{1}{4}$% of ? = 5400

?/16 = 5400

? =86400

3. 26% of x = 2564 – 2226

26% of x = 338

x= 1300

4. $2\frac{1}{4} \times 24 =$ x% of 360

9/4 $\times$ 24 = 3.6x

54 = 3.6x

x = 54/3.6 =15

5. 70% of 350 = X % of 490

70 × 350 / 100 = X × 490/ 100

70 × 350 = 490 x X

X = 70×350/ 490 = 50

6. 18 is what % of 144?

Ans = 18/144 × 100 = 100/8 =12.5%

7. $\frac{7}{20}$ is what % of $\frac{7}{10}$?

Ans = (7/20)/(7/10) × 100 = 1/2× 100 = 50%

8. 25% of what is $\frac{1}{9}$?

25% of x = 1/9

x/4 = 1/9

x = 4/9

9. 35% of what is 70 × 8

35% of x = 560

X = 560 /(35%) = 560 × 100/35 = 1600

10. 490% of 25 + 25% of 490 = ?

= 490% of 25 + 25% of 490

= 25% of 490 + 25% of 490

= 2×25% of 490

= 50% of 490

= 490/2

=245

11. 40% of 1020 + ?% = 70% of 990 + 300% of 425

408+ ? % = 693 + 1275 = 1968

? % = 1560

? = 1560 × 100 = 1,56,000

12. 600% of 1600 = ?% x 1200

9600 = 12 × ?

? = 9600/12 = 800

13. 25% of (250% of Rs.800) = ?

= 25% of (2000)

= 2000/4

= 500

14. If Y% of Y is 49, then Y is equal to__.

Y% of Y = 49

Y^2= 4900 =70

15. What is 25% of 25% equal to?

= 25% of 25%

= 0.25 × 0.25

= 0.0625

16. What is 60 percent of Rs.720?

= 60% of 720

= Rs. 432

17. What % of 500 gm is 2 Kg?

Ans = 2kg/ 500g × 100 = 2000g/ 500g × 100 = 400%

18. Two numbers are respectively 10% and 25% more than a third number. First number is how much % less than the second number?

Sol:

Let the third number = 100

So 1st number = 110% of 100 = 110

And 2nd number = 125% of 100 = 125

Hence required ans = (125-110)/125×100 = 15/125×100 = 12%

19. Two numbers are respectively 40% and 60% of the third number. What % of the first number is the second number?

Sol:

Let the third number = 100

So 1st number = 40% of 100 = 40

And 2nd number = 60% of 100 = 60

Hence required ans = 60/40×100 = 150%

20. Two numbers are respectively 20% and 30% less than a third number. What % of first number is the second number?

Sol:

Let the third number = 100

So 1st number = (100-20)% of 100 = 80

And 2nd number = (100-30)% of 100 = 70

Hence required ans = 70/80×100 = 87.5%

21. A person gives 30% of his salary to his wife and 45% of his salary to his children. Now, he is left with Rs. 5,000. Find his total salary?

Sol:

His savings = (100 – 30 – 45) % = 25% = 5000

1% =200

So total salary = 100% = Rs. 20000

22. A person gives 20% of his salary to his wife, 15% of his salary to his son and 25% of his salary to his daughter. Now, he is left with Rs. 16,000, What is his total salary?

Rs. 40,000

Sol:

His savings = (100 – 20 – 15 – 25) % = 40% = 16000

1% =400

So total salary = 100% = Rs. 40000

23. A person gives 30% of his salary to his wife and 20% of the remaining to his children. Now he is left with Rs. 28,000. What is his total salary?

Sol:

His savings = (100 – 30 –20% of 70) % =(100 – 30 – 14)% = 56% = 28000

1% =500

So total salary = 100% = Rs 50,000

24. A person gives 20% of his salary to his wife and 25% of the remaining to his children. Now, he is left with Rs. 27,000. What is his total salary?

Sol:

His savings = (100 – 20 –25% of 80) % =(100 – 20 – 20)% = 60% = 27000

1% =450

So total salary = 100% = Rs 45,000

25. In an examination, it is required to get 45% of the total marks to qualify. Ramesh got 590 marks and was declared failed by 40 marks. What are the maximum marks a student can get?

Sol:

Let total marks = M

45% of M = 590+40 = 630

M = 630 × 100 / 45 = 1400

26. In an examination, it is required to get 55% of the total marks to qualify. Suresh got 500 marks which was 5 marks more than the qualifying marks. What are the maximum marks a student can get?

Sol:

Let total marks = M

55% of M = 500 - 5 = 495

M = 495 × 100 / 55 = 900

27. Price of a toy was increased by 40% and then decreased by 25%. What was the net change in the price (in %)?

Sol:

Net change in price = a +b + (ab/100) = 40 + (-25) + {40 × (-25)/100} = 15 -10

= +5% = 5% increase

Alternate Way:

Let price of toy originally was = Rs. 100

New price after 1st increase = 140% of 100 = 140

And new price after next decrease = (100-25)% of 140 = 75% of 140 = 105

Hence net price is increased by = (105-100)/100× 100 =5%

28. Rohan gets 35% of the marks in an examination and fails by 75 marks. Sohan gets 540 marks in the same examination and fails by 60 marks. Find the maximum marks?

Sol:

Let total marks = M

35% of M + 75 = 540 + 60 = 600

35% of M = 600 – 75 = 525

M = 525 × 100 / 35 = 1500

29. A candidate who got 24% in a test failed by 45 marks. Whereas another candidates got 240 marks in the same test, which was 15 more than the passing marks, find the maximum marks?

Sol:

Let total marks = M

24% of M + 45 = 240 - 15 = 225

24% of M = 225 – 45 = 180

M = 180 × 100 / 24 = 750

30. In an examination, 25% of total students failed in Hindi, 40% failed in English and 15% in both. Find the percentage of those who passed in both the subjects.

Sol;

P(A or B) = P(A) +P(B) - P(A and B) = 25 + 40 – 15 = 50%

Hence 50% failed in atleast 1 Subject both, so rest must be cleared in at least one subject. So rest 50% students cleared in both the subjects.

31. What is the difference between $\frac{1}{4}$th of 2000 and $\frac{1}{4}$th % of 2000?

Sol:

$1/4^{th}$ of 2000 – $1/4^{th}$ % of 2000

= 500 – (500/100)

= 500 -5 = 495

32. In a college of 15,000 students, 30% of students participated in a game show. Out of all those who participated, 25% are girls. Find the number of boys participating in the show?

Sol:

Number of Students participated = 15000×30% = 4500

So number of boys = (100-25)% of 4500 = 75% of 4500 = 3375

33. In a college of 1500 students, 400 are in Arts stream, 600 are in Science stream and remaining are in Commerce stream. What is the % of students studying Science?

Sol:

Required ans = 600/1500×100 =40%

34. The sum of 25% of a number and 12% of the same number is 2590. What is 15% of that number .

Sol:

Let the number be N

25% of N+ 12% of N = 2590

37% of N = 2590

N = 7000

So 15% of the number = 15% of 7000 = 1050

35. $\frac{1}{5}$th of $\frac{2}{5}$th of $\frac{3}{5}$th of a number is 288. What is 25% of that number?

Sol:

$\frac{1}{5} \times \frac{2}{5} \times \frac{3}{5} \times \mathrm{N}$=288 or N = 288 $\times \frac{125}{6} = 48 \times 125$= 6000

So 25% of N = 25% of 6000 = 1500

36. If the height of a triangle is decreased by 10% and base of the same is increased by 20%. What will be the effect on its area?

Sol:

Area of triangle = ½ × base × height

Net change in area = a +b + (ab/100) = (-10) + (20) + {(-10) × (20)/100} = 10 -2 = +8%

= 8% increase

37. There are 1500 employees in an office. Number of females is 25% of the number of males. Find the total number of females in the office?

Sol:

Ratio of number of male to that of female = 100: 25 = 4:1

The total number of females in office = 1/(1+4) of 1500 =1/5 of 1500 = 300

38. If the length of a rectangle is increased by 10% and breadth is reduced by 20%, What will be the effect on its area?

Sol:

Area of rectangle = Length × breadth

Net change in area = a +b + (ab/100) = (10) + (-20) + {10 ×(- 20)/100} = -10 -2 = -12%

12% decrease

39. If the price of Petrol increases by 10%. By how much % a person should decrease his consumption, so that his expenditure on petrol does not increase.

Sol:

Let price of petrol increases from = 100 to 110

So Consumption will decreased from = 110 to 100

Hence decrease in consumption = (110 – 100)/110×100 = 10/110× 100

= 100/11 = 9.11% decrease

40. If the price of Petrol decreases by 10%. By how much % a person should increase his consumption, so that his expenditure on petrol does not decrease.

Sol:

Let price of petrol decreases from = 100 to 90

So Consumption will increased from = 90 to 100

Hence increase in consumption = (100 – 90)/90×100 = 10/90× 100

= 100/9 =$11\frac{1}{9}$ % decrease

41. If the price of rice increase by 20%. By how much % a person should decrease his consumption, so that his expenditure on rice does not increase.

Sol:

Let price of rice increases from = 100 to 120

So Consumption will decreased from = 120 to 100

Hence decrease in consumption = (120 – 100)/120×100 = 20/120× 100

= 100/6 = 16.66% decrease

42. If a number is added to 21 instead of being added to 12. What % of the correct value is the result?

Ans - Cannot be determined

43. If a number is added to 20 instead of being added to 120. The correct value is how much % more than the result?

Ans - Cannot be determined

44.What % of number of days in May 2008 is number of days in Feb.2008.

Sol:

In May 2008, number of days = 31

And in Feb 2008. Number of days = 29 (as it is a leap year)

So required %age = 29/31*100 = 93.55% approx..

45. By how much % number of days in a week is less than number of hours in a day?

Sol:

Number of Days in a week = 7

Number of hours in a week = 24

So required number = (24 – 7)/24×100 = 17/24×100 = 70.83%

46. A shopkeeper marks the price of a television 25% higher than the original price and them allows a discount of 25%. What profit / loss % he will get in the transaction?

Sol:

Let the original price =100

M.R.P = 125% of 100 = 125

SP = 125× (100-25)% = 125×75% = 93.75

Hence net Loss% = (100 – 93.75)/ 100×100 = 6.25% loss

Alternate way:

Let CP = 100

SP = 100 × 125% × 75% = 93.75

Hence net Loss% = (100 – 93.75)/ 100×100 = 6.25% loss

47. A shopkeeper marks the price of a mobile 20% higher than the original price and them allows a discount of 15%. What profit / loss % he will get in the transaction?

Sol:

Let CP = 100

SP = 100 × 120% × 85% =102

Hence net Loss% = (102 – 100)/ 100×100 = 2% profit

48. Shaila decided to donate 15% of her salary to an NGO. On the day of donation, she changed her mind and donated Rs. 2,475 Which was 75% of what she decided earlier. What is Shaila's total salary?

Sol:

Let total salary = 100

Donation, he thought first = 15

So ATQ, 75% of 15 = 2475

1% = 220

100% = 22000

49. There are 8200 employees in a company, out of which 65 per cent are males and 60 per cent of the males are either 40 years or older. How many females in the company are younger than 40 years?

Sol:

We cannot find data about female's age as data is given for age of males only.

50. In an Examination, Pravin scored 49 per cent marks, Rahat scored 98 per cent marks and Sonali scored 825 marks. The maximum marks of the examination are 900. What are the average marks scored by all the three students together?

Sol:

Average of all = {49% of 900+ 98% of 900+825}/3 =(441+882+825)/3= 2148/3

= 716